

U

O

W

CSIT305 Emerging Information Technologies and their Applications



UNIVERSITY
OF WOLLONGONG
AUSTRALIA

Spatial Data and Computing



UNIVERSITY
OF WOLLONGONG
AUSTRALIA

Subject Logistics

- (1) Advanced Database Overview
- (2) Spatial Data and Computing
- (3) Graph Data and Computing
- (4) Text Data and Processing

Multi-Modal Database



- (1) AI4SE: artificial intelligence for software engineering
- (2) Software Engineering in Cloud Computing I
- (3) Software Engineering in Cloud Computing II

Software



- (1) IoT and Edge Computing
- (2) Digital Twin Technology
- (3) Blockchains and Smart Contracts
- (4) Emerging Cyber-Attacks

Networking

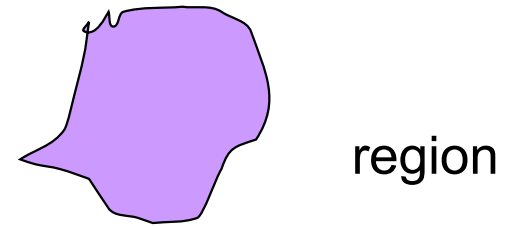


Outline

- Spatial Data
 - Spatial Query Types
 - Spatial Index Structure
 - Spatial Query Processing
- Spatial Textual Data
 - Spatial Textual Index Structure
 - Spatial Textual Query Processing
- Applications

Spatial Data

- Spatial Data Types



- Point : 2 real numbers
- Line : sequence of points
- Region : area included inside n-points